Supplement: hierarchical POMDP methods and parameters

This supplement makes the two-level construction of sec. 13.9 concrete: how location (L1) is coupled to context (L2) through context-conditioned priors, what the alternating-minimization update actually computes, and the exact parameters of the executed run. It answers the question the main section brackets — *by what mechanism does a second latent level enter the inference at all* — and thereby fixes why the hierarchy resolves context above chance while leaving location accuracy statistically unchanged.

Generative model for context-gated location inference I

The two-level POMDP implemented in `fedference.pomdp.build_hierarchical_world` couples the sentinel's 9-location L1 factor to a 2-state L2 context factor via **context-conditioned L1 priors**:

Generative model for context-gated location inference I

- **L1 (location)** — the standard 3x3 grid of `build_sentinel_world` with `n_s = 9` states and sensor acuity 0.85;
- **L2 (context)** — a binary state (`quiet` / `alert`) with a symmetric transition matrix (persistence 0.90) and an initial uniform prior;
- **L1 priors given context** — `quiet`: uniform over all 9 cells; `alert`: mass 0.60 at the center cell (the den), the residual spread uniformly.

Inference algorithm for top-down empirical priors I

`fedference.pomdp.hierarchical_infer` performs 4 passes of alternating minimization:

Inference algorithm for top-down empirical priors I

1. **L2 → L1 empirical prior**: $\widetilde{\pi}_{0,\mathrm{L1}} = \sum_c q_{\mathsf{ctx}}[c]\, \pi_{0,\mathrm{L1}|c}$ (a soft mixture of the two context-conditioned priors).
2. **L1 update**: one-step variational posterior $q_{\mathsf{loc}} = \mathrm{softmax}(\log \widetilde{\pi}_{0,\mathrm{L1}} + \log A[\mathsf{obs},\, \cdot])$.
3. **L1 → L2 marginal evidence**: $\ell_c = \log(\pi_{0,\mathrm{L1}|c}^{\top} A[\mathsf{obs},\, \cdot])$ (evidence for context c from the observed location likelihood).
4. **L2 update**: $q_{\mathsf{ctx}} = \mathrm{softmax}(\log \pi_{0,\mathrm{L2}} + \ell)$.

Inference algorithm for top-down empirical priors I

After 4 iterations the agent broadcasts both q_{loc} and q_{ctx}; the colony federates each level independently via a log-linear pool (eq. 6).

Study parameters for the hierarchical condition I

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Table 1: Study 6 hierarchical POMDP execution parameters: agent count, seeded trial budget, observation acuity, alternating-minimization iterations, and the L2/L1 state cardinalities used by the two-level condition.

Parameter	Value
Agents	4
Trials	960
Acuity	0.85
Alternating-min iterations	4
L2 context states	2
L1 location states	9
Seed	0

Study parameters for the hierarchical condition I

The executed hierarchical configuration is summarized in tbl. 1.